

Optical Fibers: Modes, Propagation, and Beam Profiles

Experiment Overview

In this experiment we study optical fibers as carriers of electromagnetic information. Using a red laser coupled into three different fibers — a multimode fiber, a 600–800 nm single-mode fiber, and a 980 nm dual-mode fiber — we investigate the transverse intensity distributions (modes) of the guided light, measure coupling efficiencies, and demonstrate the geometric phase of light.

Apparatus:

- Red laser ($\lambda \approx 633\text{--}700\text{ nm}$)
- Two front-silvered mirrors in kinematic mounts
- Fiber-optic coupler with aspheric lens on x - y translation stage
- Three optical fibers: multimode, 600–800 nm single-mode, 980 nm dual-mode
- Linear polarizer, optical power meter, CCD camera, ImageJ software

1. Physical Foundation: Total Internal Reflection

When a light ray travels from a medium of refractive index n_1 into a medium with $n_2 < n_1$, Snell's law at the interface requires:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (1)$$

Since $n_1 > n_2$, we always have $\theta_2 > \theta_1$. As θ_1 increases, θ_2 reaches 90° before θ_1 does. The critical angle θ_c is defined by the condition $\theta_2 = 90^\circ$:

$$\boxed{\sin \theta_c = \frac{n_2}{n_1}}. \quad (2)$$

For any $\theta_1 \geq \theta_c$ there is no refracted ray and all light energy is reflected back into the denser medium. This is **Total Internal Reflection (TIR)**.

An optical fiber consists of a cylindrical **core** (index n_1) surrounded by a **cladding** (index $n_2 < n_1$). Light is launched from an external medium of index n (typically air, $n = 1$) into the fiber end-face. The ray refracts at the end-face, propagates through the core, and strikes the core-cladding interface. The geometry is shown in Figure 1.

Referring to the figure, a ray enters the fiber at point A from medium n at angle α to the fiber axis. By Snell's law at the end-face:

$$n \sin \alpha = n_1 \sin \theta, \quad (3)$$

where θ is the refracted angle inside the core measured from the fiber axis. The ray travels to point B on the core-cladding interface, where it makes an angle θ_c with the normal to that interface (i.e., the normal is along Y). From the geometry at point B:

$$\theta_c = \frac{\pi}{2} - \theta, \quad \Rightarrow \quad \sin \theta_c = \cos \theta. \quad (4)$$

For TIR to occur at the core-cladding boundary, the ray must strike at exactly the critical angle or beyond. At the limiting case:

$$\sin \theta_c = \frac{n_2}{n_1} \quad \Rightarrow \quad \cos \theta = \frac{n_2}{n_1}. \quad (5)$$

Using $\cos \theta = \sqrt{1 - \sin^2 \theta}$ and substituting $\sin \theta = (n/n_1) \sin \alpha$ from Eq. (3):

$$\sqrt{1 - \frac{n^2}{n_1^2} \sin^2 \alpha} = \frac{n_2}{n_1}. \quad (6)$$

Squaring both sides and rearranging:

$$n^2 \sin^2 \alpha = n_1^2 - n_2^2. \quad (7)$$

The maximum acceptance angle $\alpha_{\max}$ is therefore:

$$n \sin \alpha_{\max} = \sqrt{n_1^2 - n_2^2}. \quad (8)$$

The **Numerical Aperture (NA)** is defined as:

$$\boxed{\text{NA} = n \sin \alpha_{\max} = \sqrt{n_1^2 - n_2^2}}. \quad (9)$$

For launch from air ($n = 1$), this simplifies to $\text{NA} = \sin \alpha_{\max}$. Only rays entering within the cone of half-angle $\alpha_{\max}$ will undergo TIR at point B and remain guided. Rays entering at steeper angles refract into the cladding at B and are lost.

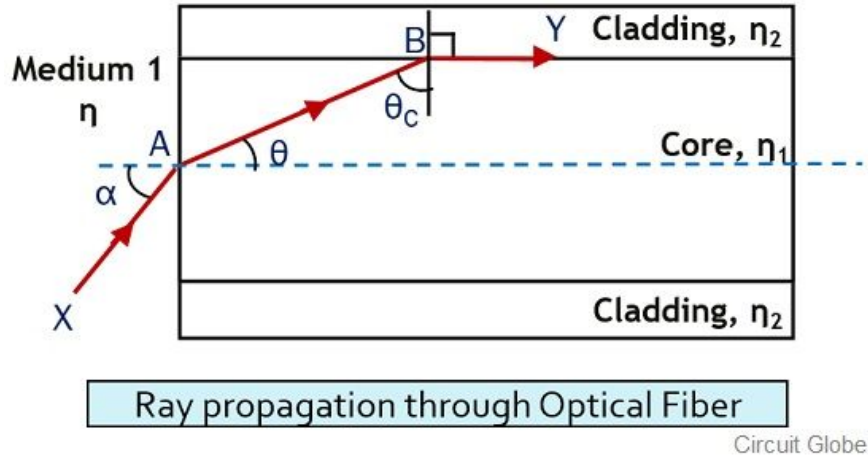


Figure 1: Ray propagation through an optical fiber. A ray enters from medium n at angle α at point A, refracts into the core (n_1) at angle θ , and strikes the core–cladding interface at point B at the critical angle θ_c . The cladding has index $n_2 < n_1$.

Key trade-off: A large NA captures more light (large $\alpha_{\max}$) but also accepts rays at steep angles θ inside the core. These steep rays travel longer path lengths and arrive at the output with different time delays, causing **modal dispersion** that limits the information bandwidth of the fiber.

2. Wave-Optic Description: Modes of a Fiber

The ray picture of TIR is intuitive but incomplete. To understand *which* field distributions are actually sustained inside the fiber, we need to solve Maxwell’s equations under the appropriate boundary conditions.

2.1 The Paraxial Wave Equation

Starting from Maxwell’s equations in a linear, isotropic medium, one obtains the vector wave equation (Helmholtz equation):

$$\nabla^2 \mathbf{E}(\mathbf{r}) + k^2 \mathbf{E}(\mathbf{r}) = 0, \quad k = \frac{n\omega}{c}. \quad (10)$$

We seek solutions that describe a beam propagating chiefly in the z -direction with a finite transverse extent. We write:

$$E(\mathbf{r}) = \mathcal{E}_0(x, y, z) e^{ikz}, \quad (11)$$

where $\mathcal{E}_0$ is a slowly varying envelope. “Slowly varying” is the **paraxial approximation**:

$$\left| \frac{\partial \mathcal{E}_0}{\partial z} \right| \ll k |\mathcal{E}_0|, \quad \left| \frac{\partial^2 \mathcal{E}_0}{\partial z^2} \right| \ll k \left| \frac{\partial \mathcal{E}_0}{\partial z} \right|. \quad (12)$$

Substituting into the Helmholtz equation and dropping the small second derivative in z , we arrive at the **Paraxial Wave Equation (PWE)**:

$$\boxed{\nabla_T^2 \mathcal{E}_0 + 2ik \frac{\partial \mathcal{E}_0}{\partial z} = 0}, \quad \nabla_T^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}. \quad (13)$$

This equation is mathematically identical in structure to the time-dependent Schrödinger equation of the quantum harmonic oscillator, with the role of time played by the propagation coordinate z . This analogy is the key to finding all solutions.

2.2 The Fundamental Gaussian Mode

We try the ansatz:

$$\mathcal{E}_0(\mathbf{r}) = A \exp\left(\frac{ik(x^2 + y^2)}{2q(z)}\right) e^{ip(z)}, \quad (14)$$

where $q(z)$ is the complex beam parameter and $p(z)$ accounts for phase. Substituting into Eq. (13) and solving the resulting ordinary differential equations yields the **Gaussian beam solution**:

$$E(\mathbf{r}) = \frac{E_0}{\sqrt{1 + (z/z_R)^2}} \exp\left(-\frac{x^2 + y^2}{w(z)^2}\right) \exp\left(ikz + \frac{ik(x^2 + y^2)}{2R(z)} - i \arctan \frac{z}{z_R}\right), \quad (15)$$

characterized by the following parameters:

Beam waist	w_0 : minimum spot size, occurring at $z = 0$.
Rayleigh range	$z_R = \pi w_0^2/\lambda$: distance over which the beam area doubles.
Beam radius	$w(z) = w_0 \sqrt{1 + (z/z_R)^2}$: $1/e$ intensity radius at position z .
Wavefront radius	$R(z) = z [1 + (z_R/z)^2]$: radius of curvature of phase fronts.
Far-field divergence	$\theta = \lambda/(\pi w_0)$: half-angle at which the beam spreads.
Gouy phase	$\phi_G(z) = \arctan(z/z_R)$: additional axial phase shift.

2.3 Hermite–Gaussian Modes

The PWE has a complete, orthonormal family of solutions beyond the fundamental Gaussian. Using the separable ansatz:

$$\mathcal{E}_0(\mathbf{r}) = A g\left(\frac{x}{w(z)}\right) h\left(\frac{y}{w(z)}\right) e^{iP(z)} \exp\left(\frac{ik(x^2 + y^2)}{2q(z)}\right), \quad (16)$$

and substituting into the PWE gives two independent Hermite differential equations. The solutions are the **Hermite–Gaussian (HG) modes** HG_{mn} :

$$E_{mn}(x, y, z) = A \frac{w_0}{w(z)} H_m\left(\frac{\sqrt{2}x}{w(z)}\right) H_n\left(\frac{\sqrt{2}y}{w(z)}\right) \times \exp\left(-\frac{x^2 + y^2}{w(z)^2}\right) \exp\left[i\left(kz - (m + n + 1) \arctan \frac{z}{z_R}\right)\right] \exp\left(\frac{ik(x^2 + y^2)}{2R(z)}\right), \quad (17)$$

where H_m is the Hermite polynomial of order m . The indices m and n label the number of nodes in the x - and y -directions respectively. These modes are also called **TEM_{mn}** modes (Transverse ElectroMagnetic).

Analogy with the quantum harmonic oscillator: Just as the quantum number n labels the eigenstates of the harmonic oscillator — with the ground state ($n = 0$) being a Gaussian probability density — the indices m, n label the eigenmodes of the PWE. The HG_{00} (TEM_{00}) mode is a pure Gaussian intensity profile and is the fiber’s “ground state.” Higher-order modes have larger transverse extents, analogous to excited states of the oscillator.

The intensity profile of HG_{mn} along the x -axis (at beam center) behaves as:

$$I_m(x) \propto H_m^2\left(\frac{\sqrt{2}x}{w}\right) \exp\left(-\frac{2x^2}{w^2}\right). \quad (18)$$

In particular:

- $m = 0$: $H_0 = 1$, so $I \propto e^{-2x^2/w^2}$ — a single Gaussian lobe.
- $m = 1$: $H_1(u) = 2u$, so $I \propto x^2 e^{-2x^2/w^2}$ — a double-lobed profile with a node at the centre.

3. Single-Mode vs. Multimode Fibers: The V-Number

Whether a fiber supports one or many modes depends on its **normalized frequency**, also called the **V-number**:

$$V = \frac{2\pi a}{\lambda} \text{NA} = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}, \quad (19)$$

where a is the core radius and λ is the free-space wavelength. The physical meaning of V is the ratio of the core size to the wavelength, weighted by the refractive-index contrast.

3.1 Single-Mode Condition

A step-index fiber guides only the fundamental HG_{00} mode when:

$$V < 2.405. \quad (20)$$

The value 2.405 is the first zero of the Bessel function J_0 , which arises from the exact cylindrical-symmetry solution of Maxwell’s equations in the fiber. Below this threshold, higher-order modes are cut off — they cannot form standing waves across the core and decay evanescently into the cladding.

3.2 Number of Guided Modes

For $V \gg 1$ (multimode regime), the approximate number of guided modes is:

$$N \approx \frac{V^2}{2}. \quad (21)$$

A typical multimode fiber with $a = 25 \mu\text{m}$, $\text{NA} = 0.22$, and $\lambda = 633 \text{ nm}$ has $V \approx 55$ and supports ~ 1500 modes simultaneously.

Fiber type	Core diameter	V-number condition
Single-mode (600–800 nm)	$\sim 8 \mu\text{m}$	$V < 2.405$ (only HG_{00})
Dual-mode (980 nm fiber)	$\sim 10\text{--}12 \mu\text{m}$	$2.405 < V < 3.832$ ($\text{HG}_{00} + \text{HG}_{01/10}$)
Multimode	$\gtrsim 50 \mu\text{m}$	$V \gg 1$ (hundreds of modes)

Why the 980 nm fiber supports two modes with a red laser: The V-number depends on the ratio a/λ . Although the fiber is *labelled* for 980 nm (at which it is single-mode), when illuminated with a 633–700 nm laser the effective V-number increases (since λ is smaller), pushing the fiber into the dual-mode regime. The HG_{01} mode now fits inside the larger (relative to wavelength) core.

4. Relation Between HG and Laguerre–Gaussian Modes

Hermite–Gaussian modes are the natural solutions in Cartesian coordinates, while **Laguerre–Gaussian (LG) modes** arise in cylindrical coordinates. Both sets form complete, orthonormal bases for paraxial beams, and any mode in one basis can be expressed as a linear combination of modes in the other.

The most important relation for this experiment is:

$$\text{HG}_{00} = \text{LG}_0^{+1} + \text{LG}_0^{-1}, \quad (22)$$

and more generally:

$$\text{LG}_p^\ell = \text{HG} \cdot e^{\pm i\ell\phi}, \quad (23)$$

which implies:

$$\text{LG}_0^{+1} + \text{LG}_0^{-1} = \text{HG} \cdot \cos \phi. \quad (24)$$

When the incidence angle ϕ of the beam on the input mirror changes, $\cos \phi$ varies, causing the superposition of LG modes — and hence the apparent HG output pattern — to *rotate*. This rotation of the output beam profile when mirror screws are adjusted is a direct manifestation of the **geometric (Berry’s) phase** of the photon’s polarization state.

5. Far-Field Analysis and Beam Waist Measurement

In the characterization of single-mode fibers (SMFs), the transverse field distribution of the fundamental LP_{01} mode is conventionally approximated by a Gaussian profile. Let (x', y') denote the coordinates on the fiber output facet. The near-field amplitude is:

$$\psi(x', y') = A \exp\left(-\frac{x'^2 + y'^2}{w^2}\right) \quad (25)$$

where w is the Gaussian beam waist.

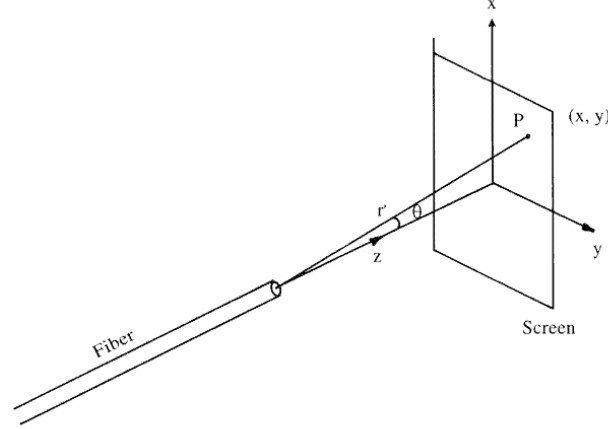


Figure 2: Geometry for far-field imaging of the fiber output.

The far-field amplitude u is determined by the Fraunhofer diffraction integral, which relates the far field to the angular distribution via a Fourier transform:

$$u = C \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi(x', y') e^{ik_0(lx' + my')} dx' dy' \quad (26)$$

Here, $l = \sin \theta \cos \phi$ and $m = \sin \theta \sin \phi$ are direction cosines. Due to cylindrical symmetry, we evaluate the distribution along the x -axis ($\phi = 0$), simplifying the kernel to $e^{ik_0 x' \sin \theta}$. Substituting Eq. (??) into the above:

$$u(\theta) = AC \int_{-\infty}^{+\infty} e^{-y'^2/w^2} dy' \int_{-\infty}^{+\infty} e^{-x'^2/w^2} e^{ik_0 x' \sin \theta} dx' \quad (27)$$

Using the Gaussian integral identity $\int_{-\infty}^{+\infty} e^{-px^2 + qx} dx = \sqrt{\pi/p} e^{q^2/4p}$, both integrals evaluate to:

$$u(\theta) = AC (\sqrt{\pi} w) \left(\sqrt{\pi} w \exp \left[-\frac{k_0^2 \sin^2 \theta}{4/w^2} \right] \right) = AC \pi w^2 \exp \left(-\frac{k_0^2 w^2 \sin^2 \theta}{4} \right) \quad (28)$$

The corresponding far-field intensity distribution is:

$$\boxed{I(\theta) = I(0) \exp \left(-\frac{k_0^2 w^2 \sin^2 \theta}{2} \right)} \quad (29)$$

This is a Gaussian in $\sin \theta$, confirming that the far-field pattern of a single-mode fiber is also Gaussian — a direct consequence of the Fourier-transform relationship between near and far fields.

The angle θ_e at which the far-field intensity falls to $1/e$ of the on-axis value satisfies:

$$\boxed{\sin \theta_e = \frac{2}{k_0 w} = \frac{\lambda_0}{\pi w}} \quad (30)$$

Rearranging, the beam waist is directly obtained from the measured θ_e :

$$w = \frac{\lambda_0}{\pi \sin \theta_e} \quad (31)$$

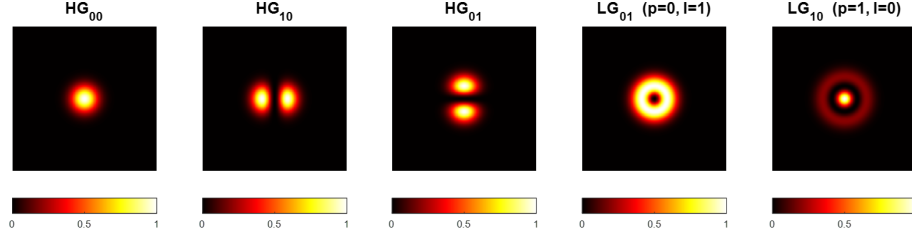


Figure 3: Two-dimensional normalized intensity maps of HG_{00} , HG_{10} , HG_{01} , LG_{01} , and LG_{10} modes computed analytically. Color scale runs from 0 (black) to 1 (white).

This provides a clean, non-invasive method to determine the mode field diameter of a single-mode fiber without direct access to the near-field. In practice, one measures the far-field intensity profile on a screen at a known distance L from the fiber tip, fits a Gaussian to the profile, and extracts θ_e from the $1/e$ half-width r_e via $\sin \theta_e \approx \tan \theta_e = r_e/L$ (valid for small angles).

6. Transverse Intensity Profiles of Optical Beam Modes

The intensity of the HG_{mn} mode is:

$$I_{mn}(x, y) \propto H_m^2\left(\frac{\sqrt{2}x}{w}\right) H_n^2\left(\frac{\sqrt{2}y}{w}\right) \exp\left(-\frac{2(x^2 + y^2)}{w^2}\right). \quad (32)$$

HG_{00} gives a single Gaussian spot; HG_{10} and HG_{01} give double-lobed profiles with nodes at $x = 0$ and $y = 0$ respectively. The intensity of the $LG_{p\ell}$ mode is:

$$I_{p\ell}(r) \propto \left(\frac{\sqrt{2}r}{w}\right)^{2|\ell|} \left[L_p^{|\ell|}\left(\frac{2r^2}{w^2}\right)\right]^2 \exp\left(-\frac{2r^2}{w^2}\right), \quad (33)$$

where $L_p^{|\ell|}$ is the associated Laguerre polynomial, p is the radial index, and ℓ is the azimuthal index. LG_{01} ($p = 0$, $\ell = 1$, $L_0^1 = 1$) produces a donut ring with a phase singularity at the axis; LG_{10} ($p = 1$, $\ell = 0$, $L_1^0 = 1 - 2r^2/w^2$) produces a central spot surrounded by a concentric outer ring [Fig. 3 and 4].

7. Geometric Phase in Optical Fibers

The geometric phase (also called Berry's phase) is an additional phase acquired by a quantum (or classical wave) state when it undergoes a *cyclic* evolution in parameter space. For photons in an optical fiber, two distinct mechanisms produce geometric phases:

(a) Pancharatnam–Berry phase (polarization evolution): When the polarization state of a photon traces a closed loop on the Poincaré sphere, it acquires a geometric phase equal to half the solid angle Ω enclosed by that loop:

$$\Phi_g = -\frac{\Omega}{2}. \quad (34)$$

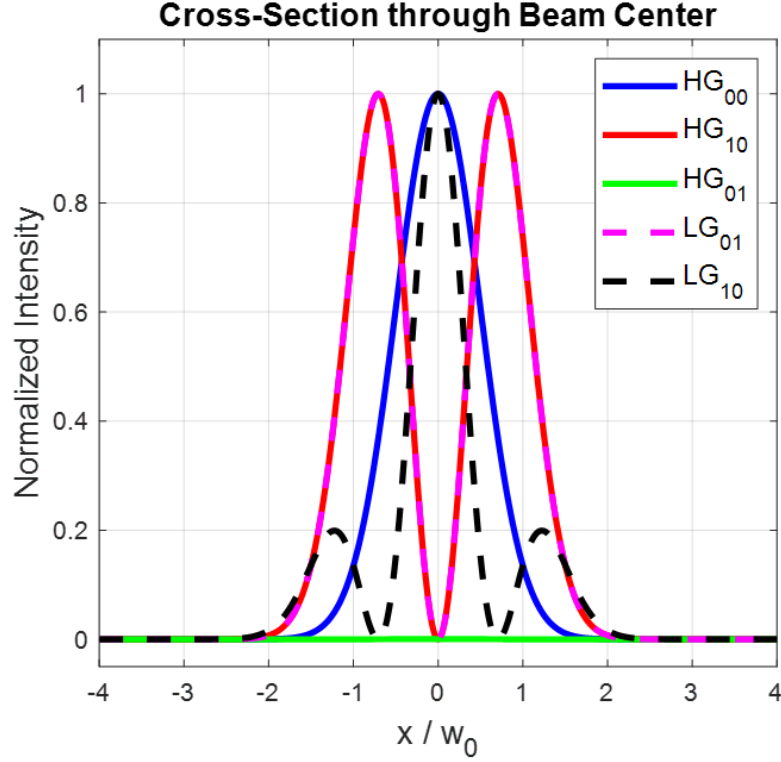


Figure 4: Horizontal cross-sections through the beam centre for all five modes. The structural differences — single peak (HG_{00}), double lobe (HG_{10} , HG_{01}), off-axis ring (LG_{01}), and bullseye (LG_{10}) — are directly visible.

(b) Mode evolution in helical fibers: In a helically wound fiber, the propagation direction of the fiber traces a cone. The geometric phase acquired due to the fiber's spatial geometry is:

$$\Phi_g = \oint (1 - \cos \theta) d\phi, \quad (35)$$

where θ and ϕ describe the local fiber orientation. In the original Tomita–Chiao experiment (1986), linearly polarized light propagating through a helically coiled single-mode fiber acquired a rotation of its polarization plane by exactly the solid angle subtended by the helix axis on the unit sphere — confirming Berry's prediction.

Observation in this experiment: When the mirror screws are adjusted, the input beam's transverse mode structure changes slightly. Because HG modes decompose into LG modes carrying orbital angular momentum $\pm \ell \hbar$, changing the input phase ϕ shifts the relative phase of the LG_0^{+1} and LG_0^{-1} components, rotating the output intensity pattern. This is the mode-space geometric phase and is observed for both the multimode fiber and the dual-mode fiber.

8. Information Quality and Quantity in Fibers

8.1 Quantity: Power and Coupling Efficiency

The *quantity* of information carried is represented by the transmitted optical power. The **coupling efficiency** η is:

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%. \quad (36)$$

Sources of power loss include:

- **Reflection at the fiber face** (Fresnel loss, $\sim 4\%$ per interface for glass–air).
- **Numerical aperture mismatch**: light outside the acceptance cone is not guided.
- **Misalignment**: the focused spot may not coincide with the fiber core.
- **Absorption and Rayleigh scattering** along the fiber length.
- **Bending loss** (microbending and macrobending).

In general, a multimode fiber accepts more light (larger core, larger NA) so $\eta_{\text{MM}} > \eta_{\text{DM}} > \eta_{\text{SM}}$.

8.2 Quality: Modal Dispersion and Beam Profile

The *quality* of information relates to how faithfully the transverse spatial structure of the beam is preserved. Each guided mode travels at a slightly different group velocity (different propagation constant β_{mn}). In a multimode fiber, hundreds of modes are simultaneously excited and interfere at the output, producing a complex **speckle pattern** that bears little resemblance to the input Gaussian beam. This is **modal dispersion** — the primary limitation of multimode fibers for telecommunications.

A single-mode fiber eliminates modal dispersion entirely: only HG_{00} propagates, and the output is always a clean Gaussian, regardless of fiber length. This is why all long-haul optical communications use single-mode fiber.

Property	Single-mode	Dual-mode	Multimode
Output profile	Clean Gaussian	Two HG modes	Speckle pattern
Modal dispersion	None	Minimal	Severe
Coupling efficiency	Low ($\sim 55\text{--}67\%$)	Moderate ($\sim 90\%$)	High ($\sim 93\%$)
Bandwidth	Very high	High	Low
Best application	Telecom, internet	Mode-division multiplexing	Spectroscopy, short links

9. Bibliographic Notes and Further Reading

Important remark on the theory presented in this manual: The Hermite–Gaussian modes derived in Section 2 are solutions of the *free-space* paraxial wave equation. They describe laser beams propagating in unbounded homogeneous media. The modes of an optical fiber are fundamentally different in origin — they arise from the *boundary conditions* imposed by the core–cladding interface, not from free-space propagation. This section clarifies the distinction and points the reader to sources for a rigorous treatment.

9.1 Free-Space Modes vs. Fiber Modes: A Critical Distinction

In free space, the paraxial wave equation admits the Hermite–Gaussian (HG) family as its complete orthonormal set of solutions. These modes are self-similar upon propagation — they retain their functional form and only their width $w(z)$ and phase evolve with z . There is no boundary, no interface, and no confinement mechanism; the Gaussian profile arises purely from the structure of the differential equation itself.

In an optical fiber the situation is qualitatively different. The electromagnetic field must satisfy Maxwell's equations *separately* in the core (index n_1) and the cladding (index n_2), with the boundary conditions at the core–cladding interface requiring continuity of the tangential components of $\mathbf{E}$ and $\mathbf{H}$. For a step-index fiber of core radius a , this leads to the **characteristic equation** (also called the *eigenvalue equation* of the fiber):

$$\left[\frac{J'_m(ua)}{u J_m(ua)} \right] + \left[\frac{K'_m(wa)}{w K_m(wa)} \right] = m\beta \left[\frac{1}{u^2 a^2} + \frac{1}{w^2 a^2} \right] \left[\frac{n_1^2}{u^2 a^2} + \frac{n_2^2}{w^2 a^2} \right]^{1/2}, \quad (37)$$

where J_m is the Bessel function of the first kind of order m , K_m is the modified Bessel function of the second kind, β is the propagation constant, and

$$u^2 = n_1^2 k_0^2 - \beta^2, \quad w^2 = \beta^2 - n_2^2 k_0^2. \quad (38)$$

The parameter u governs the oscillatory field in the core, and w governs the evanescent decay in the cladding. The discrete allowed values of β — found by solving the characteristic equation — label the **guided modes** of the fiber, designated HE_{mn} , EH_{mn} , TE_{0n} , and TM_{0n} depending on their polarization character.

Under the **weakly-guiding approximation** ($n_1 \approx n_2$, i.e., small refractive-index contrast), the vector modes degenerate in pairs and can be combined into the **linearly polarized (LP) modes** $\text{LP}_{\ell m}$, whose transverse field profiles are:

$$\Psi_{\ell m}(r, \phi) \propto \begin{cases} J_\ell(ur) \cos(\ell\phi), & r \leq a \quad (\text{core}), \\ K_\ell(wr) \cos(\ell\phi), & r > a \quad (\text{cladding}). \end{cases} \quad (39)$$

The fundamental mode LP_{01} ($\ell = 0$, first zero of the characteristic equation) has a *nearly* Gaussian transverse profile, which is why the Gaussian approximation used in Section 2 and Section 4 works well in practice for single-mode fibers. However, the true profile is $J_0(ur)$ in the core and $K_0(wr)$ in the cladding, not a pure Gaussian. The Gaussian is an approximation whose accuracy depends on the V-number; it improves as $V \rightarrow 2.405^-$.

Summary of the distinction:

- **Free-space HG modes:** arise from the symmetry of the paraxial wave equation in unbounded homogeneous space. No boundary conditions involved.
- **Fiber LP modes:** arise from matching electromagnetic boundary conditions at the core-cladding interface. The transverse profile is a Bessel function, not a Hermite polynomial.
- **The connection:** for a single-mode fiber near cutoff, $LP_{01} \approx HG_{00}$ (Gaussian). This is why the free-space formalism gives quantitatively correct results in many practical situations, even though it is not the rigorous fiber-mode theory.

9.2 References for Further Study

The following texts and articles are recommended for students wishing to go beyond the scope of this manual. They are arranged from introductory to advanced.

Textbooks

- [1] A. Yariv and P. Yeh, *Photonics: Optical Electronics in Modern Communications*, 6th ed. Oxford University Press, 2007.
Chapters 3 and 4 give a thorough derivation of fiber modes, the characteristic equation, and LP modes under the weakly-guiding approximation. Highly recommended as the primary reference.
- [2] J. M. Senior, *Optical Fiber Communications: Principles and Practice*, 3rd ed. Pearson, 2009.
Accessible treatment of fiber modes, numerical aperture, V-number, single-mode condition, and dispersion. Good starting point for beginners.
- [3] K. Okamoto, *Fundamentals of Optical Waveguides*, 2nd ed. Academic Press, 2006.
Rigorous and comprehensive. Covers the full vector mode theory, weakly-guiding approximation, LP modes, and dispersion in detail. Suitable for advanced readers.
- [4] G. P. Agrawal, *Nonlinear Fiber Optics*, 6th ed. Academic Press, 2019.
Chapter 2 provides an excellent self-contained derivation of the single-mode fiber field equations, the LP mode approximation, and the Gaussian beam waist approximation. Indispensable for understanding pulse propagation and dispersion.
- [5] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, 3rd ed. Wiley, 2019.
Chapter 9 (Fiber Optics) gives a pedagogically clear derivation of the guided modes, with good figures. Chapters 3 and 4 cover Gaussian beam optics and the paraxial approximation. Excellent for building physical intuition alongside the mathematics.
- [6] A. W. Snyder and J. D. Love, *Optical Waveguide Theory*. Chapman and Hall, 1983.
The definitive reference for waveguide mode theory. Contains the complete derivation of the characteristic equation, all mode types, and the weakly-guiding approximation. Advanced level; use as a reference rather than a tutorial.

Original Research Articles

- [7] D. Gloge, “Weakly guiding fibers,” *Applied Optics*, vol. 10, no. 10, pp. 2252–2258, 1971.
The original paper introducing the LP mode approximation. Essential reading to

understand where the weakly-guiding formalism comes from.

- [8] A. Tomita and R. Y. Chiao, “Observation of Berry’s topological phase by use of an optical fiber,” *Physical Review Letters*, vol. 57, no. 8, pp. 937–940, 1986.
The landmark experiment demonstrating the geometric (Berry’s) phase of photons in a helically wound single-mode fiber. Directly relevant to the geometric phase demonstration in this experiment.
- [9] M. V. Berry, “Quantal phase factors accompanying adiabatic changes,” *Proceedings of the Royal Society of London A*, vol. 392, pp. 45–57, 1984.
Berry’s original theoretical paper predicting the geometric phase. Foundational reading for understanding the physical origin of the phase observed in the mirror-rotation demonstration.
- [10] Zhou Yi, Wu Zhen-hua, and Ge Mo-lin, “Geometric phase in optical fiber,” *Chinese Physics Letters*, vol. 16, no. 5, pp. 316–317, 1999.
Extends Berry’s phase analysis to fibers with linear birefringence. Directly related to the theory behind this experiment.
- [11] M. A. Bandres and J. C. Gutiérrez-Vega, “Ince–Gaussian beams,” *Optics Letters*, vol. 29, no. 2, pp. 144–146, 2004.
For students interested in the broader family of paraxial beam solutions beyond HG and LG modes.

Online Resources

- [12] *RP Photonics Encyclopedia* — <https://www.rp-photonics.com/encyclopedia.html>
Freely accessible, peer-reviewed encyclopedia of photonics. Articles on fiber modes, numerical aperture, Gaussian beams, and geometric phase are particularly relevant.
- [13] *MIT OpenCourseWare: 6.007 Electromagnetic Energy*
<https://ocw.mit.edu>
Lecture notes on waveguide modes and fiber optics at an undergraduate level. Good supplementary reading before tackling the graduate-level texts.

10. Experimental Procedure

10.1 Optical Alignment: The Near–Far Method

Alignment is the most critical skill in this experiment. The goal is to make the laser beam **perfectly parallel to the optical bench** (breadboard), so that it enters the aspheric lens at normal incidence and focuses directly into the fiber core.

Two mirrors are necessary because a single mirror can only adjust the beam direction, not simultaneously fix both the position and direction at an arbitrary point. With two mirrors, any beam can be made parallel to the bench at any height using the *near–far* technique:

1. **Establish target height:** measure the center height of the fiber coupler from the breadboard. This is your target beam height.
2. **Near-near correction:** place a ruler or card *close to* the far mirror (Mirror 2). If the beam height differs from target, adjust the **near mirror** (Mirror 1) vertical screw.
3. **Far-far correction:** move the card *far from* Mirror 2. If the height differs,

- adjust the **far mirror** (Mirror 2) vertical screw.
4. Iterate steps 2 and 3 until the beam height is constant everywhere. Repeat for horizontal position.
 5. Insert the fiber mount. The beam should enter the lens center. If not, adjust Mirror 2 only (do not re-do the parallelism alignment).

The cardinal rule: “Near-near, Far-far.” When correcting the beam at a *near* point (close to the far mirror), adjust the *near* mirror. When correcting at a *far* point, adjust the *far* mirror.

10.2 Coupling and Beam Walking

Once the beam is parallel and entering the lens center, use the multimode fiber first (easiest to couple due to its large core). Maximize the output power using the x - y translation stage of the fiber mount. Then switch to the single-mode fiber and perform **beam walking**:

1. Reduce power by moving the horizontal screw of Mirror 2. Recover (or exceed) the power using Mirror 1’s horizontal screw. This walks the beam to better angular alignment.
2. Repeat with vertical screws.
3. Alternate between mirrors iteratively until coupling is maximized.
4. **Do not change one mirror’s horizontal screw and the other’s vertical screw simultaneously** — this scrambles the alignment.

10.3 Measurements

10.3.1 Power (Quantity) Measurement

Measure the input laser power P_{in} with the power meter placed at the fiber input (before the coupler). Then measure the output power P_{out} for each of the three fibers. Record at least three readings and compute:

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%, \quad \Delta\eta = \eta \left(\frac{\Delta P_{\text{in}}}{P_{\text{in}}} + \frac{\Delta P_{\text{out}}}{P_{\text{out}}} \right). \quad (40)$$

10.3.2 Beam Profile (Quality) Measurement

Project the fiber output onto a white card and image it with the CCD camera into ImageJ. Avoid saturation. Use the *straight line* selection tool to extract a pixel grey-value profile across the beam. Then in Python or Origin, fit the profile:

For the single-mode fiber (HG₀₀ mode):

$$I(x) = I_0 + A \exp\left(-\frac{(x-b)^2}{2c^2}\right). \quad (41)$$

For the dual-mode fiber (HG₁₀ mode):

$$I(x) = I_0 + \frac{A}{w} (x-b)^2 \exp\left(-\frac{2(x-c)^2}{w^2}\right). \quad (42)$$

Record the goodness-of-fit R^2 and compare with the theoretical profile.

10.3.3 Polarizer Experiment (Dual-mode Fiber)

The two modes of the 980 nm fiber (HG_{00} and HG_{10}) are orthogonally polarized because the Hermite polynomials are orthonormal eigenfunctions. Insert the polarizer after the fiber output and rotate it:

- At one orientation, you should see the Gaussian (HG_{00}) profile.
- At 90° rotation, the double-lobed (HG_{10}) profile appears.
- Intermediate angles give a superposition.

This demonstrates that orthogonal modes can be separated by a linear polarizer — a key principle in **mode-division multiplexing** for high-capacity fiber communications.

10.3.4 Geometric Phase Demonstration

With the 980 nm fiber or the multimode fiber, observe the output pattern on the camera. Slowly rotate a screw of one mirror (changing the angle of incidence on the mirror). Observe how the output intensity pattern rotates. This rotation is a consequence of the geometric phase: as $\cos \phi$ changes, the relative phase of the LG components constituting the HG output mode changes, causing the pattern to rotate.

11. Data Analysis Guide

11.1 Extracting the Beam Profile in ImageJ

1. Open the CCD image in ImageJ (File → Open).
2. Check the image is not saturated: Analyze → Histogram. If the count at the maximum pixel value (255 for 8-bit) is large, the image is saturated.
3. Draw a horizontal line across the beam center using the *Straight Line* tool.
4. Go to Analyze → Plot Profile to get the grey-value vs. pixel-position data.
5. Export as CSV for fitting in Python or Origin.

11.2 Fitting in Python

```
import numpy as np
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt

# Load data
data = np.loadtxt("profile.csv", delimiter=",", skiprows=1)
x, I = data[:,0], data[:,1]

# HG00 fit
def gaussian(x, I0, A, b, c):
    return I0 + A * np.exp(-(x-b)**2 / (2*c**2))

popt, pcov = curve_fit(gaussian, x, I, p0=[5, 50, 500, 100])
perr = np.sqrt(np.diag(pcov))

# R-squared
```

```

residuals = I - gaussian(x, *popt)
ss_res = np.sum(residuals**2)
ss_tot = np.sum((I - np.mean(I))**2)
R2 = 1 - ss_res / ss_tot

plt.plot(x, I, 'k.', label='Data')
plt.plot(x, gaussian(x, *popt), 'r-', label=f'Fit,  $R^2={R2:.4f}$ ')
plt.xlabel("Pixel position"); plt.ylabel("Grey value")
plt.legend(); plt.tight_layout(); plt.savefig("fit.pdf")

```

12. Summary and Expected Results

Expected observations:

- **Multimode fiber:** Speckled, irregular output with many intensity peaks. Pixel profile is irregular and cannot be fit by a single Gaussian. Highest coupling efficiency ($\eta_{MM} \sim 75\text{--}93\%$).
- **Single-mode fiber (600–800 nm):** Clean, smooth Gaussian output. Profile fits $I(x) = A \exp(-(x - b)^2/2c^2)$ with $R^2 > 0.99$. Lower coupling efficiency ($\eta_{SM} \sim 55\text{--}67\%$) because the small core is harder to align.
- **Dual-mode fiber (980 nm):** Without polarizer, output shows a broad profile (superposition of HG_{00} and HG_{10}). With polarizer at one orientation: Gaussian. At 90° : double-lobed profile fitting the HG_{10} function.
- **Geometric phase:** Rotating a mirror screw by a small amount causes the output intensity pattern (especially for the multimode fiber) to rotate, demonstrating the mode-space geometric phase.
- **Coupling efficiency ranking:** $\eta_{MM} > \eta_{DM} > \eta_{SM}$, consistent with the idea that multimode fibers accept a wider cone of light.

Conclusion: Single-mode fibers preserve the *quality* of information (beam profile) at the cost of lower coupling efficiency. Multimode fibers transmit more *quantity* (power) but scramble the spatial mode structure. The dual-mode fiber is an intermediate case allowing two independent spatial channels. These trade-offs directly determine the applications of each fiber type in real-world photonics and communications systems.